

Chapter 35- IP Phones, Voice VLANs & QoS



1. Introduction to IP Phones (VoIP Basics)

Traditional Telephony

- Traditional phones use:
 - **PSTN (Public Switched Telephone Network)**
 - Also called **POTS (Plain Old Telephone System)**

 This is the old system where voice travels over **dedicated telephone lines**

IP Phones (Modern Communication)

- IP Phones use **VoIP (Voice Over IP)**
- Voice travels over an **IP network (Internet or LAN)**

 Means:

- Calls are sent as **data packets**  over network
-

How IP Phones Connect

- IP Phones connect to switches like normal devices
- BUT they are **special devices** 

Internal 3-Port Switch inside IP Phone

An IP Phone has **3 ports**:

1.  **Uplink Port** → Connects to Switch
2.  **Downlink Port** → Connects to PC
3.  **Internal Port** → Connects to Phone itself

👉 This allows:

- **PC + Phone = Single switch port usage**
-

💡 Real-Life Example

Imagine a **desk setup**:

- One cable from wall → Phone
 - PC connects to phone
 - 👉 Saves ports + clean wiring
-

🔗 2. Voice VLAN Concept

🚨 Problem

- PC traffic + Voice traffic both go through same cable

👉 But:

- Voice needs **high priority**
 - Data traffic can tolerate delay
-

✅ Solution: Voice VLAN

- Separate traffic into **2 VLANs**:
 - 💻 Data VLAN → PC traffic
 - 📞 Voice VLAN → IP Phone traffic
-

📦 Traffic Behavior

- PC Traffic → **UNTAGGED**
 - Phone Traffic → **TAGGED with VLAN ID**
-

? Exam Point

👉 Why not use trunk mode?

✓ Answer:

- Because:
 - **Access VLAN + Voice VLAN = Special combination**
 - Works in **access mode only**
 - Not a normal trunk

⚙️ Configuration

```
interface g0/0
switchport mode access
switchport access vlan 10
switchport voice vlan 11
```

🔍 Verification Commands

```
show interfaces g0/0 switchport
show interfaces g0/0 trunk
```

💡 Real-Life Example

- Voice VLAN = **VIP lane on highway** 🚗
 - Data VLAN = **Normal traffic lane**
 - 👉 VIP (voice) always gets priority
-

⚡ 3. Power Over Ethernet (PoE)

🔌 What is PoE?

- PoE allows:
 - **Power + Data over same Ethernet cable**
-

⚙️ Components

- **PSE (Power Sourcing Equipment)** → Switch
 - **PD (Powered Device)** → IP Phone, Camera, AP
-

⚙️ Working Process

1. Switch sends **low power signal**
 2. Detects device requirement
 3. Supplies required power only
-

⚠️ Why Important?

- Prevents:
 - 🔥 Device damage
 - ⚡ Overcurrent issues
-

💡 Real-Life Example

Like:

- Charging phone with **smart charger**
 - 👉 Gives only required voltage
-

⚠️ 4. PoE Risks & Protection (Power Policing)

🔔 Problem

- Device may draw **more power than required**
- Can damage:
 - Device 🔥
 - Switch port ⚠️

🛡️ Solution: Power Policing

⚙️ Default Behavior

power inline police

👉 Same as:

power inline police action err-disable

✓ If over power:

- Port goes **err-disabled**
- Syslog message generated

🔄 Recovery

shutdown

no shutdown

🔍 Verification

show power inline police g0/0

Alternative Mode

power inline police action log

✓ Behavior:

- Does NOT shutdown port
- Logs message
- Restarts interface

5. PoE Standards Table

Name	Standard	Power	Wire Pairs
Cisco ILP Non-standard		7W	2
PoE	802.3af	15W	2
PoE+	802.3at	30W	2
UPOE	802.3bt	60W	4
UPOE+	802.3bt	100W	4

6. Introduction to QoS (Quality of Service)

Old Network Design

- Voice → PSTN
- Data → IP Network

👉 No competition → No QoS needed

Modern Networks (Converged Networks)

- Voice 
- Video 
- Data 

 All share same network

Problem

- All traffic competes for bandwidth
-

Solution: QoS

 QoS = Tools to:

- Control traffic
 - Prioritize important packets
-

7. QoS Key Parameters

1. Bandwidth

- Capacity of link (bps)

 QoS can:

- Reserve bandwidth for voice
-

2. Delay

- Time to reach destination

Types:

- One-way delay
 - Two-way delay
-

3. Jitter

- Variation in delay between packets

👉 IP phones use:

- **Jitter buffer** to smooth audio
-

4. Packet Loss

- % of packets not delivered

Causes:

- Bad cables ❌
 - Network congestion ❌
-

8. QoS Standards (VERY IMPORTANT ⚠️)

Metric	Acceptable Value
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One-way Delay	≤ 150 ms
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Jitter	≤ 30 ms
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Packet Loss	$\leq 1\%$
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If exceeded:

- Voice becomes:
 - Broken 
 - Delayed 
 - Robotic 
-

Real-Life Example

- Video call lagging 
 That is:
 - High delay
 - High jitter
 - Packet loss
-

9. Key Commands Summary

Voice VLAN

```
switchport mode access  
switchport access vlan 10  
switchport voice vlan 11
```

PoE

```
power inline police  
power inline police action log  
show power inline police
```

QoS (Conceptual in this lecture)

- No heavy config yet
 - Focus on:
 - Delay
 - Jitter
 - Loss
 - Bandwidth
-

10. Important Exam Points

- ✓ IP Phone has **3-port switch**
 - ✓ Voice VLAN works with **access mode**
 - ✓ PC traffic = **untagged**
 - ✓ Voice traffic = **tagged**
 - ✓ PoE supplies **power + data**
 - ✓ QoS controls **traffic priority**
-

Final Understanding (Simple Way)

 Think of network like a **road system**:

-  Data traffic → Normal vehicles
-  Voice traffic → Ambulance

 QoS ensures:

- Ambulance always gets priority 
-



SUMMARY



IP Phones & VoIP

- Old system → **PSTN / POTS** 📞
- New system → **VoIP (Voice over IP)** 🌐
- Voice is sent as **data packets over network**



IP Phone has **3-port switch**:

-  Uplink → Switch
-  Downlink → PC
-  Internal → Phone

✓ Saves switch ports + cables



Voice VLAN

- Used to **separate Voice & Data traffic**

Traffic Type	Behavior
Data (PC)	Untagged
Voice	Tagged



✓ Voice VLAN gives **priority to voice traffic**



PoE (Power over Ethernet)

- Sends **Power + Data on same cable**

Device Type Role



PSE Switch



PD IP Phone

✓ Smart power delivery (no damage)

Power Policing

- Prevents **over-power damage**
- Default action:
 **Err-disable port**

QoS (Quality of Service)

- Controls **network traffic priority**

Key Parameters:

-  Bandwidth
-  Delay
-  Jitter
-  Packet Loss

Important Values:

- Delay ≤ 150 ms
- Jitter ≤ 30 ms
- Loss $\leq 1\%$

CONCLUSION

 Modern networks are **converged networks** (Voice + Video + Data)

✓ IP Phones + Voice VLAN ensure:

- Proper traffic separation
- Better call quality

✓ PoE simplifies infrastructure:

- No extra power cables

✓ QoS ensures:

- Voice traffic gets **highest priority**

💡 Final Understanding:

👉 Without QoS → Voice becomes **laggy, broken, robotic**

👉 With QoS → Smooth communication 🚀

Q & A

? Q1. What is VoIP?

✓ Answer:

VoIP (Voice over Internet Protocol) is a technology that allows voice communication over IP networks instead of traditional telephone lines.

👉 Voice is converted into digital packets and transmitted over network.

✓ Advantage:

- Cost-effective
- Flexible
- Scalable

? Q2. Difference between PSTN and VoIP?

Feature	PSTN	VoIP
Network	Telephone line	IP network
Cost	High	Low
Flexibility	Low	High

? Q3. How does an IP Phone work?

✓ Answer:

An IP phone:

- Converts voice → digital packets
- Sends over network
- Uses protocols like SIP

👉 It connects to switch like a normal device.

? Q4. What are the ports in IP Phone?

✓ Answer:

IP Phone has 3 ports:

- Uplink → Switch
- Downlink → PC
- Internal → Phone

👉 This allows PC + Phone on one cable.

? Q5. What is Voice VLAN?

✓ Answer:

Voice VLAN is a VLAN specifically used to carry voice traffic.

- ✓ It separates voice and data traffic
 - ✓ Improves performance and quality
-

? Q6. Why Voice VLAN is needed?

✓ Answer:

Because:

- Voice requires **low delay & jitter**
- Data traffic can tolerate delay

👉 So voice traffic is prioritized.

? Q7. Difference between Data VLAN and Voice VLAN?

Feature	Data VLAN	Voice VLAN
Traffic	PC data	Voice
Tagging	Untagged	Tagged
Priority	Normal	High

? Q8. Why voice VLAN is not trunk mode?

✓ Answer:

Because:

- It uses **special access mode**
- Allows:
 - Untagged (PC)
 - Tagged (Phone)

👉 Not a normal trunk link.

? Q9. Voice VLAN configuration?

✓ Answer:

```
interface g0/0
switchport mode access
switchport access vlan 10
switchport voice vlan 11
```

? Q10. What is PoE?

✓ Answer:

PoE allows devices to receive **power + data over Ethernet cable**.

? Q11. What are PSE and PD?

✓ Answer:

- PSE → Power Source (Switch)
 - PD → Powered Device (IP Phone, Camera)
-

? Q12. How PoE works?

✓ Answer:

1. Switch sends low power
2. Detects device
3. Supplies required power

👉 Prevents damage

? Q13. What is Power Policing?

✓ Answer:

Power policing protects devices from over-power.

👉 If device consumes more power:

- Port goes **err-disabled**

? Q14. Difference between “police” and “log”?

Mode Action

police Shutdown port

log Only log

? Q15. PoE Standards?

Standard Power

802.3af 15W

802.3at 30W

802.3bt 60–100W

? Q16. What is QoS?

✓ Answer:

QoS (Quality of Service) is a technique used to prioritize important traffic like voice and video.

? Q17. Why QoS is required?

✓ Answer:

Because:

- Multiple traffic types share same network
 - Voice needs priority
-

? Q18. What are QoS parameters?

✓ Answer:

- Bandwidth
 - Delay
 - Jitter
 - Packet Loss
-

? Q19. What is Jitter?

✓ Answer:

Jitter is variation in packet delay.

👉 High jitter = bad voice quality

? Q20. What is acceptable delay?

✓ Answer:

- ≤ 150 ms
-

? Q21. What happens if QoS not applied?

✓ Answer:

- Voice becomes:
 - Broken 🔊
 - Delayed ⌚
 - Robotic 🤖

🔧 ◆ PRACTICAL QUESTIONS

? Q22. How to verify Voice VLAN?

✓ Answer:

show interfaces g0/0 switchport

? Q23. How to check PoE status?

✓ Answer:

show power inline

? Q24. How to recover err-disabled port?

✓ Answer:

shutdown
no shutdown

? Q25. Troubleshooting: Phone not working?

✓ Answer:

Check:

- ✓ VLAN config
 - ✓ PoE power
 - ✓ Cable
 - ✓ QoS
-

? Q26. Why voice is breaking during calls?

✓ Answer:

Possible reasons:

- High delay
 - High jitter
 - Packet loss
 - No QoS
-

? Q27. Explain full packet flow?

✓ Answer:

1. Phone generates voice
 2. Tagged with VLAN
 3. Switch forwards
 4. QoS prioritizes
 5. Destination receives
-

One Liner

 “Explain complete Voice VLAN + QoS setup”

Best Answer:

- Configure access + voice VLAN
- Ensure PoE enabled
- Apply QoS for priority
- Verify with commands

 Say this confidently → You’ll look **PRO level** 

FINAL TAKEAWAY

- ✓ IP Phone = Smart network device
- ✓ Voice VLAN = Traffic separation
- ✓ PoE = Power + Data
- ✓ QoS = Priority control

 Together they ensure:

 **Clear, fast, and reliable communication**
